exeta

Agentic Coding with OpenCode

Q-SUMMIT 2026

Building software with AI agents that plan, act, and observe

50-minute workshop

Agenda

Time	Section
00:00–05:00	What is Agentic Coding?
05:00–15:00	OpenCode: Architecture & Core Concepts
15:00–25:00	MCP Servers & Permissions
25:00–40:00	Live Demos (Obsidian Plugin + CLI)
40:00–50:00	Patterns, Pitfalls & Q&A

Goal: Leave able to run OpenCode on a real project today.

Current standing

GitHub

Claude

- Big enterprise solutions: Copilot / Claude
- They are agentic tools to work with LLM models in a smart and complex way
- It just works™
- Not open source → proprietary models
- Not flexible, if you want them to be. #Apple

What is Agentic Coding?

Traditional coding assistants → **autocomplete / chat**

Agentic coding → **plan** → **act*** → **observe** → **repeat**

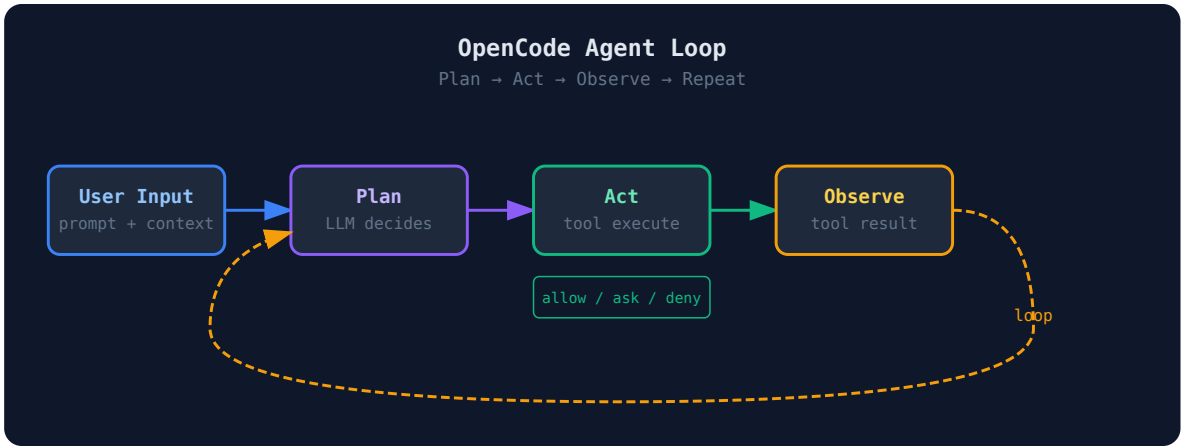
The key differences is the **Act** phase:

- **Autonomous multi-step execution** — the agent decides the next action
- **Tool use** — reads files, runs bash, searches the web
- **Feedback loops** — observes output and self-corrects
- **Configurable trust boundaries** — you decide what's auto-approved
- **Delegate work to further agents** – offload the tasks to spezialized agents

> "You are now a junior developer on your team — except it ships code instantly"

What Is an Agent?

An AI agent receives a goal, proposes a plan of steps, executes permitted tools, observes results, and iterates — subject to permissions and human oversight.



⚠️ Agents can still produce incorrect results — human review remains essential.

Why Multiple Agents?

One agent with a broad scope often compromises. Multiple scoped agents give better safety, modularity, and parallelism.

Single Agent	Multi-Agent (scoped)
One context, one role, all permissions	Each agent has a narrow scope
Generalist reasoning	Specialist prompts & scopes
One trust boundary for everything	Least-privilege per role

Example team: @lead orchestrates → @analyst researches → @dev codes → @critic challenges → @qa validates

⚠️ "Specialist" means different prompts and permission scopes, not autonomous judgment — keep humans in the loop.

OpenCode

OpenCode is an open-source agent orchestrations tool built for the terminal.

```
# Install
curl -fsSL https://opencode.ai/install | bash
# or
npm install -g opencode-ai
```

```
# Run in your project
opencode
```

- Open-source (GitHub: anomalyco/opencode)
- Works with any LLM provider (Anthropic, OpenAI, Bedrock, local models)
- TUI, desktop app, IDE extension, CLI — same config everywhere
- First-class multi-agent orchestration built in

OpenCode Architecture

```
┌──────────────────────────────────────┐
│            User Interface            │
│  TUI  │  IDE Extension  │  Web       │
└──────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────┐
│          OpenCode Core               │
│  Agent Loop  │  Tool Router          │
│  Permissions │  Context Manager      │
└──────────────────────────────────────┘
        │                  │
        ▼                  ▼
┌──────────┐        ┌──────────────┐
│  LLM     │        │    Tools     │
│Provider  │        │bash│read│MCP  │
└──────────┘        └──────────────┘
```

Config files: `opencode.json` (global or per-project) + `AGENTS.md`

The Agent Loop

```
User Message
     │
     ▼
                  tool call
 ┌──────────┐   ──────────►   ┌──────────┐
 │ LLM Plan │                 │ Tool Execute │
 │ (think)  │   ◄──────────   │ (act)    │
 └──────────┘   tool result   └──────────┘
     │                            │
     ▼                            ▼
 Next step?                   Permission check
 Done? Ask?                   allow / ask / deny
```

Modes:

- **Build** (default) — full tool access
- **Plan** — read-only, edits require approval → switch with `Tab`

Built-in Tools

Tool	Purpose
`bash`	Run shell commands (`npm install` , `git status`)
`read write edit`	Read and manipulate file contents (line-range aware)
`skill`	Load a `SKILL.md` instruction set
`todowrite`	Manage task lists in-session
`delegate`	Delegates a task to a subagent and reads the response
`question`	Ask user for input mid-task
`tool_use`	Reads and uses MCP tool commands from active servers

MCP Servers — What & Why

MCP = Model Context Protocol

A standard for connecting LLMs to external tools and services.

Why it matters:

- Extend OpenCode with **any external API** (Sentry, Jira, GitHub, databases)
- Tools become first-class citizens alongside built-ins
- The LLM uses them the same way — just another tool call
- Context cost: each MCP server adds tokens → use selectively

```
OpenCode ◀──── stdio (local) ────▶ MCP Process
OpenCode ◀──── HTTPS (remote) ────▶ MCP Server
```

MCP Config: Local Server

```
{
  "$schema": "https://opencode.ai/config.json",
  "mcp": {
    "filesystem": {
      "type": "local",
      "command": ["npx", "-y", "@modelcontextprotocol/server-
                  "/path/to/allowed"],
      "enabled": true,
      "environment": {
        "DEBUG": "1"
      }
    }
  }
}
```

MCP Config: Remote + OAuth

```
{
  "$schema": "https://opencode.ai/config.json",
  "mcp": {
    "sentry": {
      "type": "remote",
      "url": "https://mcp.sentry.dev/mcp",
      "oauth": {}
    },
    "context7": {
      "type": "remote",
      "url": "https://mcp.context7.com/mcp",
      "headers": {
        "CONTEXT7_API_KEY": "{env:CONTEXT7_API_KEY}"
      }
    }
  }
}
```

Permissions — Why They Matter

Without permissions: the agent can do anything.

- Delete production config
- Push to main without review
- Read `.env` files with secrets
- Execute `rm -rf` variants

OpenCode permission model: 3 outcomes per tool:

- `"allow"` — run without prompt
- `"ask"` — show UI dialog (once / always / reject)
- `"deny"` — block entirely

Default: most tools `allow`, `.env` files `deny`, `doom_loop` and `external_directory` → `ask`

Permission Config — Real Examples

```
{
  "$schema": "https://opencode.ai/config.json",
  "permission": {
    "*": "ask",
    "bash": {
      "*": "ask",
      "git *": "allow",
      "npm *": "allow",
      "rm *": "deny",
      "grep *": "allow"
    },
    "edit": {
      "*": "deny",
      "packages/web/src/content/docs/*.mdx": "allow"
    },
    "read": {
      "*": "allow",
      "*.env.*": "deny",
      "*.env.example": "allow"
(…)
```

Rules evaluated **last-match-wins**. Put `"*"` first, specifics after.

Configuring your agent team

Before agents can collaborate effectively, you need to tell them who they are and what they're allowed to do.

- **What this section covers:** AGENTS.md, agent files, permissions
- **Why it matters:** configuration is how you enforce roles, scope, and safety boundaries

AGENTS.md — Project Rules

Rule files injected as plain text into every agent session's system prompt:

File	When active	Best for
`AGENTS.md` in project root	Sessions in that directory (or subdirs)	Build commands, architecture, conventions
`~/.config/opencode/AGENTS.md`	Every session, all projects	Personal preferences (not committed to Git)

Both files are **combined** — project rules stack on top of global rules.

Bootstrap with `/init` — scans your repo, may ask clarifying questions, then creates *or improves* an existing `AGENTS.md` in place. Learn more: /init docs

AGENTS.md — Project Rules

```
# Obsidian Plugin Demo

## Off-limits
- Do not modify `manifest.json` version field during demos
- Do not run `npm publish`

## Available Agents
- **@coder** — TypeScript / Obsidian API specialist
```

Example from this workshop's own `.opencode/AGENTS.md`

> Commit `AGENTS.md` to Git — it's shared team knowledge, not personal config.

> ⚠️ AGENTS.md is plain text — agents follow it as instructions, not as enforced rules. Permissions in `opencode.json` are the true enforcement layer.

Agent Files — Configuration Shape

Markdown agents live in:

- Global: `~/.config/opencode/agents/review.md`
- Project: `.opencode/agents/review.md`

```
---
description: Code review without making edits
mode: subagent
model: anthropic/claude-sonnet-4-20250514
temperature: 0.1
permission:
  edit: deny
  bash:
    "git diff": allow
    "grep *": allow
  webfetch: deny
---
You are a code reviewer. Focus on:
- Security and input validation
- Performance implications
- Code clarity and maintainability

Provide constructive feedback. Do not make changes.
```

Agent Config: JSON Style

```
{
  "$schema": "https://opencode.ai/config.json",
  "agent": {
    "build": {
      "mode": "primary",
      "model": "anthropic/claude-sonnet-4-20250514",
      "permission": {
        "bash": {
          "*": "ask",
          "git status *": "allow",
          "npm run *": "allow"
        }
      }
    },
    "code-reviewer": {
      "description": "Reviews code for best practices",
      "mode": "subagent",
      "permission": {
        "edit": "deny",
        "bash": "deny"
      }
    }
```

Built-in Agents

Primary (cycle with Tab):

- `build` — full access (default)
- `plan` — edits/bash require ask

Subagents (@ mention or auto):

- `general` — full access, parallel tasks
- `explore` — read-only, fast codebase search

Hidden system agents:

- `compaction` — auto-compacts long contexts
- `title` — generates session titles
- `summary` — session summaries

What you'll build

In the next two demos you'll go from zero to a working agentic workflow.

- **Goal:** scaffold and analyze real code with OpenCode — no manual edits
- **Step 1:** Use Plan mode to review an architecture before touching files
- **Step 2:** Switch to Build mode and watch the agent execute the plan
- **Step 3:** Run a non-interactive analysis pass suitable for CI pipelines

Expected outcome: a scaffolded Obsidian plugin and a structured refactoring report, produced entirely by the agent under your permission rules.

Demo 1: Obsidian Plugin Scaffold

Goal: Use OpenCode to scaffold a new Obsidian plugin from scratch.

Setup:

```
cd workshop-agentic/demo-repo/obsidian-plugin
opencode
```

Steps we'll follow:

1. Switch to **Plan mode** (Tab) — ask for architecture overview
2. Review the plan together
3. Switch to **Build mode** (Tab) — execute
4. Watch agent: read existing examples → write plugin boilerplate → run `npm install`
5. Use `/undo` to revert one change and re-prompt

What to observe:

- Tool calls streamed live (bash, write, read)

Demo 1: Walkthrough Notes

```
Plan output example:
1. Create manifest.json with plugin metadata
2. Create main.ts with Plugin class extending Plugin
3. Create styles.css for status bar styling
4. Create package.json with esbuild config
5. Run npm install

Build execution:
[write] manifest.json          → auto-allow (new file)
[write] main.ts                → auto-allow (new file)
[bash] npm install             → PERMISSION PROMPT
  > once | always | reject
[bash] npm run build           → PERMISSION PROMPT
```

Key observations:

- Agent reads `obsidian` npm package types to understand the API
- Glob searches for existing examples before writing
- Context7 MCP could provide live Obsidian API docs

Demo 2: CLI Plan Mode on Existing Repo

Goal: Analyze a messy CLI repo, get a refactoring plan — no changes made.

```
cd workshop-agentic/demo-repo/cli-fallback
opencode run --agent plan "Analyze this CLI codebase.
Identify the top 3 refactoring opportunities.
Output a structured plan with effort estimates."
```

This uses:

- `opencode run` — single-shot, non-interactive
- `--agent plan` — forces Plan mode (no edits)
- Stdout output suitable for CI pipelines

Demo 2: Walkthrough

```
# Non-interactive analysis
opencode run --agent plan \
  "Find all TODO comments and create a prioritized issue list"

# Pipe output to a file
opencode run --agent plan \
  "Summarize the public API surface of this module" > api-summary.md

# With custom AGENTS.md context
OPENCODE_CONFIG=./opencode.ci.json opencode run \
  "Check for any hardcoded credentials or API keys"
```

Real-world CI pattern:

```
# .github/workflows/ai-review.yml
- name: AI Code Review
  run: |
    opencode run --agent plan \
      "Review this PR diff for security issues. Be concise."
```

Skills — Reusable Instruction Sets

Skills are named `SKILL.md` files loaded on-demand by agents.

```
.opencode/skills/git-release/SKILL.md
~/.config/opencode/skills/deploy-checklist/SKILL.md
```

```
---
name: git-release
description: Create consistent releases and changelogs
license: MIT
compatibility: opencode
---

## What I do
- Draft release notes from merged PRs
- Propose a version bump
- Provide a copy-pasteable `gh release create` command

## When to use me
Use when preparing a tagged release.
```

Agent sees available skills in tool description → loads on demand via `skill({ name: "git-release" })`

Real-World Patterns

Pattern 1: Trust Tiers

```
read-only tier:    plan agent, explore subagent
restricted tier:   git allow, npm allow, rm deny
full-trust tier:   only in dev, never in CI
```

Pattern 2: Per-Agent MCP Scoping

```
{ "tools": { "my-mcp*": false },
  "agent": { "research": { "tools": { "my-mcp*": true } } } }
```

→ MCP tools only available to the agent that needs them

Pattern 3: Skills for Team Workflows

```
.opencode/skills/release-checklist/SKILL.md
.opencode/skills/pr-template/SKILL.md
.opencode/skills/security-review/SKILL.md
```

→ Shared, versioned, Git-committed team knowledge

Pattern 4: AGENTS.md as Living Docs Update after every significant architecture decision.

Pitfalls & Anti-Patterns

❌ "Allow everything" in shared environments

```
{ "permission": "allow" }  // NEVER in CI or shared machines
```

❌ **Secrets in config files**

```
{ "headers": { "Authorization": "Bearer sk-abc123" } }  // bad
{ "headers": { "Authorization": "Bearer {env:MY_KEY}" } } // good
```

❌ **Ignoring `doom_loop` warnings** Three identical tool calls = agent is stuck. Intervene, don't ignore.

❌ **Giant AGENTS.md** The whole file is injected every session. Keep it focused, use Skills for detail.

❌ **No `/init` before starting** Without AGENTS.md, the agent guesses your conventions. Run `/init` first.

✅ **Always start with Plan mode for unfamiliar codebases**

Q&A — Discussion Prompts

For the audience:

1. Where in your current workflow would an agentic step add the most value?
2. What's the riskiest thing you'd worry about giving an agent access to?
3. Has anyone tried MCP servers beyond the built-ins? What worked?

Resources:

- Docs: https://opencode.ai/docs
- GitHub: https://github.com/anomalyco/opencode
- Discord: https://opencode.ai/discord
- This workshop: `workshop-agentic/` in the Q-Summit repo

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Thank You

Q-SUMMIT 2026

Building software with AI agents that plan, act, and observe

Let's build with agents

Workshop materials:

```
git clone <q-summit-repo>
cd workshop-agentic
# Follow README.md
```

OpenCode: `curl -fsSL https://opencode.ai/install | bash`

Questions, feedback, follow-up: use the Q-Summit Discord channel **#agentic-coding**